L4 baselines

L4: missing baselines for Koblitz decomposition (2026-11-14) [pcc:unverified]

Question. The owner said three baselines were never compared: (a) fixing m-1 factor-base points and taking gcd(f, L_V mod f); (b) libFES Gray-code exhaustive search; (c) polynomial-method solvers (Dinur 2021, BKW 2019).

Prior work, audited by repository grep. research-kb and the lead atlas are unavailable in this container.

- b. is **closed by prior work**. FES is implemented (`mq_fes.rs`, `fes-f2`, `fes-f2-wide`) and compared with `inherited-f4`, the default Koblitz descent engine, on identical seeded targets: `research/inherited_f4_next_20261112` and `research/fes_then_batched_20261112`. A validated run is frozen but blocked (`fes_k13_validated_20261112`).
- c. is **closed by prior work**. `research/blindspots_20261018/03-polynomial-method` implements Dinur's MultParityCount and LPTWY, with real calls and a validated counted model. Its model puts the crossover with FES at N = 80 (Dinur, 2^71 bits of memory) and N = 128 (LPTWY), with none under a realistic memory cap. That study's verdict was INCONCLUSIVE on a sensitivity control.
- a. **exists** as `src/cryptanalysis/semaev_decomp.rs` (S4 pairs-and-solve with gcd(quartic, L_V mod quartic)). It had never been counted against the descent engines on the same instances. The owner is right on (a) only in that narrow sense.

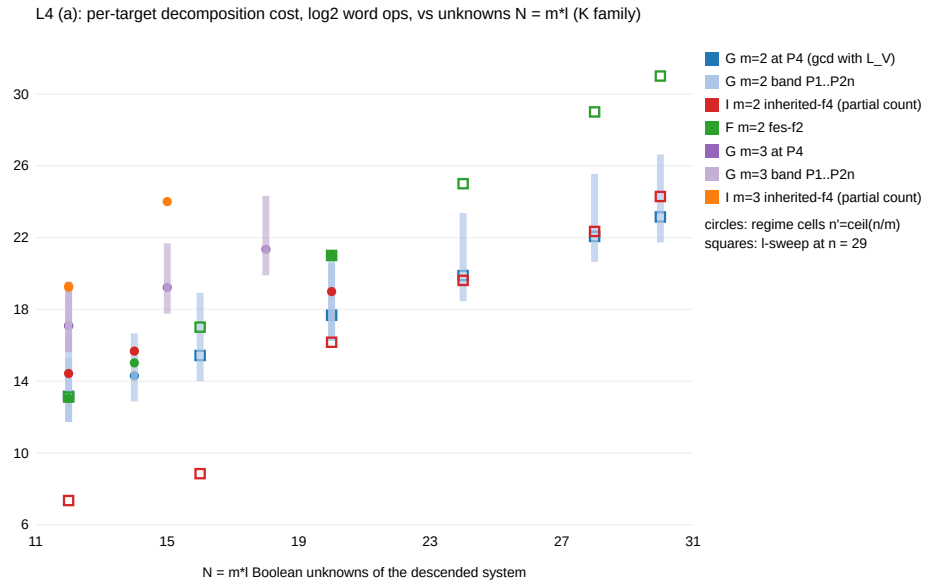
What was measured here, for (a). A lane-local Rust harness (`src/l4bench.rs`) counts the field operations of the literal gcd-with-L_V oracle. It runs on exactly the targets that the repository's `ic descent` drew and solved with `inherited-f4` and `fes-f2`: 13 cells on the Koblitz family K, n from 11 to 29, m = 2 and 3, 136 targets. Solution sets agree exactly with every engine and with brute force. A wrong-subspace negative control differs on 45 targets.

Findings.

- m = 2: G costs about 2^l (2l + 8) multiplications. It is below FES at all three prices from l = 10, and 231× below at P4 at l = 15. Against `inherited-f4` it has the same exponent (log2 slope 1.11 against 1.10 per unit of l in the regime cells). The ratio is a flat factor of 2.2 to 2.6 at P4, and its sign flips inside the pricing band. On overdetermined cells (n = 29, l ≤ 10), `inherited-f4` is up to 1000× cheaper than G.

- m = 3: G is |V|^2 quartic gcds, about 110 mul each. It beats `inherited-f4` at every price at l = 5 (27× at P4). At l = 6, `inherited-f4` did not finish in 120 s.
- G is 7 to 14× (m = 2) and 300 to 1500× (m = 3) above the pair-table / subtract oracles that the current IC baseline rows actually use.

Verdict. The frozen rule is ambiguous. Its literal clause gives IMPROVES as a stage result against `inherited-f4` at m = 3. Against the baseline rows' own oracle the result is NEGATIVE. In either reading, **no improvement of the index-calculus baseline**, and no change to the exponent.



cost vs unknowns